

52-mavzu. Teylor formulasi

52-ma'ruza

Reja

- 1^o. Ko'p o'zgaruvchili funksiyaning Teylor formulasi.
- 2^o. Xususiy hollar. Aralash hosilaning tengligi haqida teorema.

1^o. Ko'p o'zgaruvchili funksiyaning Teylor formulasi.

Aytaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funksiya ochiq $E \subset R^m$ to'plamda berilgan bo'lib, $U_\delta(x^0) \subset E$ bo'lsin, bunda $x^0 = (x_1^0, x_2^0, \dots, x_m^0)$ va $\delta > 0$. Ravshanki,

$$\forall x = (x_1, x_2, \dots, x_m) \in U_\delta(x^0), \quad x^0 = (x_1^0, x_2^0, \dots, x_m^0)$$

nuqtalarni birlashtiruvchi to'g'ri chiziq kesmasi

$$A = \{x_1^0 + t(x_1 - x_1^0), x_2^0 + t(x_2 - x_2^0), \dots, x_m^0 + t(x_m - x_m^0); 0 \leq t \leq 1\}$$

shu $U_\delta(x^0)$ ga tegishli bo'ladi.

Faraz qilaylik, $f(x_1, x_2, \dots, x_m)$ funksiya $U_\delta(x^0)$ to'plamda $(n+1)$ marta differensiallanuvchi bo'lsin. Bu funksiyaning A to'plamda qarasak, $[0,1]$ segmentda aniqlangan ushbu

$$F(t) = f(x_1^0 + t(x_1 - x_1^0), x_2^0 + t(x_2 - x_2^0), \dots, x_m^0 + t(x_m - x_m^0))$$

funksiyaga ega bo'lamiz. $F(t)$ funksiya $[0,1]$ da hosilaga ega bo'lib,

$$\begin{aligned} F'(t) &= \frac{\partial f}{\partial x_1} (x_1 - x_1^0) + \frac{\partial f}{\partial x_2} (x_2 - x_2^0) + \dots + \frac{\partial f}{\partial x_m} (x_m - x_m^0) = \\ &= \left(\frac{\partial}{\partial x_1} (x_1 - x_1^0) + \frac{\partial}{\partial x_2} (x_2 - x_2^0) + \dots + \frac{\partial}{\partial x_m} (x_m - x_m^0) \right) f \end{aligned}$$

bo'ladi, bunda $f(x)$ funksiyaning barcha xususiy hosilalari

$$(x_1^0 + t(x_1 - x_1^0), x_2^0 + t(x_2 - x_2^0), \dots, x_m^0 + t(x_m - x_m^0)) \quad (4)$$

nuqtada hisoblangan.

Umuman, hosil qilingan $F(t)$ funksiya k -tartibli ($k=1, 2, \dots, n+1$) hosilalarga ega va u

$$F^{(k)}(t) = \left(\frac{\partial}{\partial x_1} (x_1 - x_1^0) + \frac{\partial}{\partial x_2} (x_2 - x_2^0) + \dots + \frac{\partial}{\partial x_m} (x_m - x_m^0) \right)^k f$$

ga teng, bundagi barcha xususiy hosilalar (4) nuqtada hisoblangan. Bu munosabatning to'g'riligi matematik induksiya usuli yordamida isbotlanadi.

Shunday qilib, $F(t)$ funksiya $F'(t), F''(t), \dots, F^{(n+1)}(t)$ hosilalarga ega bo'ladi. Taylor formulasiga ko'ra (qaralsin, 24-ma'ruza) t_0 nuqtada ($0 \leq t_0 \leq 1$)

$$F(t) = F(t_0) + F'(t_0)(t - t_0) + \frac{1}{2!} F''(t_0)(t - t_0)^2 + \dots + \frac{1}{n!} F^{(n)}(t_0)(t - t_0)^n + \frac{1}{(n+1)!} F^{(n+1)}(c)(t - t_0)^{n+1} \quad (5)$$

bo'ladi, bunda $c = t_0 + \theta(t - t_0)$, $0 < \theta < 1$. Bu tenglikda $t_0 = 0, t = 1$ deyilsa, unda

$$F(1) = F(0) + \frac{1}{1!} F'(0) + \frac{1}{2!} F''(0) + \dots + \frac{1}{n!} F^{(n)}(0) + \frac{1}{(n+1)!} F^{(n+1)}(\theta)$$

bo'lishi kelib chiqadi.

Ayni paytda,

$$\begin{aligned} F(0) &= f(x_1^0, x_2^0, \dots, x_m^0), \\ F(1) &= f(x_1, x_2, \dots, x_m), \end{aligned} \quad (6)$$

$$F^{(k)}(0) = \left(\frac{\partial}{\partial x_1} (x_1 - x_1^0) + \frac{\partial}{\partial x_2} (x_2 - x_2^0) + \dots + \frac{\partial}{\partial x_m} (x_m - x_m^0) \right)^k f$$

(bunda f funksiyaning barcha xususiy hosilalari $(x_1^0, x_2^0, \dots, x_m^0)$ nuqtada hisoblangan) bo'lishini e'tiborga olsak, u holda (5) va (6) tengliklardan ushbu

$$f(x_1, x_2, \dots, x_m) = f(x_1^0, x_2^0, \dots, x_m^0) +$$

$$\begin{aligned}
& + \sum_{k=1}^n \frac{1}{k!} \left(\frac{\partial}{\partial x_1} (x_1 - x_1^0) + \frac{\partial}{\partial x_2} (x_2 - x_2^0) + \dots + \frac{\partial}{\partial x_m} (x_m - x_m^0) \right)^k f(x_1^0, x_2^0, \dots, x_m^0) + \\
& + \frac{1}{(n+1)!} \left(\frac{\partial}{\partial x_1} (x_1 - x_1^0) + \frac{\partial}{\partial x_2} (x_2 - x_2^0) + \dots + \frac{\partial}{\partial x_m} (x_m - x_m^0) \right)^{n+1} \times \\
& \times f(x_1^0 + \theta(x_1 - x_1^0), x_2^0 + \theta(x_2 - x_2^0), \dots, x_m^0 + \theta(x_m - x_m^0))
\end{aligned}$$

$(0 < \theta < 1)$ tenglikka kelimiz. Bu ko'p o'zgaruvchili $f(x_1, x_2, \dots, x_m)$ funksiyaning Lagranj ko'rinishidagi qoldiq hadli Teylor formulasi deyiladi.

2°. Xususiy hollar. Aralash hosilaning tengligi haqida teorema.

$m=1$ bo'lsin. bu holda $u=f(x) \quad (x \in R, u \in R)$ funksiyaning yuqori tartibli hosila va differensiallariga kelimiz. Ular 23-ma'ruzada batafsil bayon etilgan.

$m=2$ bo'lganda $u=f(x, y) \quad ((x, y) \in R^2, u \in R)$ ikki o'zgaruvchili funksiya bo'lib. Uning ikkinchi tartibli xususiy hosilalari (ular 4 ta bo'ladi) quyidagicha bo'ladi:

$$\frac{\partial^2 f(x, y)}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f(x, y)}{\partial x} \right),$$

$$\frac{\partial^2 f(x, y)}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f(x, y)}{\partial y} \right),$$

$$\frac{\partial^2 f(x, y)}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f(x, y)}{\partial x} \right),$$

$$\frac{\partial^2 f(x, y)}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f(x, y)}{\partial y} \right).$$

1-misol. Ushbu

$$f(x, y) = \arctg \frac{x}{y} \quad (y \neq 0)$$

funksiyaning ikkinchi tartibli xususiy hosilalari topilsin.

◀ Ravshaki,

$$\frac{\partial f}{\partial x} = \frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{1}{y} = \frac{y}{x^2 + y^2}, \quad \frac{\partial f}{\partial y} = \frac{1}{1 + \frac{x^2}{y^2}} \cdot \left(-\frac{x}{y^2} \right) = -\frac{x}{x^2 + y^2}$$

bo'ldi.

Endi ta'rifdan foydalanib berilgan funksiyaning ikkinchi tartibli xususiy hosilalarini topamiz:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{y}{x^2 + y^2} \right) = -\frac{2xy}{(x^2 + y^2)^2},$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right) = \frac{x^2 - y^2}{(x^2 + y^2)^2},$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(-\frac{x}{x^2 + y^2} \right) = \frac{x^2 - y^2}{(x^2 + y^2)^2},$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(-\frac{x}{x^2 + y^2} \right) = \frac{2xy}{(x^2 + y^2)^2}. \quad \blacktriangleright$$

2-misol. Ushbu

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & \text{agar } x^2 + y^2 > 0 \text{ бўлса,} \\ 0, & \text{agar } x^2 + y^2 = 0 \text{ бўлса} \end{cases}$$

funksiyaning $(0,0)$ nuqtadagi aralash hosilalari topilsin.

Aytaylik, $(x, y) \neq (0,0)$ bo'lsin. Bu holda

$$\frac{\partial f}{\partial x} = y \left(\frac{x^2 - y^2}{x^2 + y^2} + \frac{4x^2 y^2}{(x^2 + y^2)^2} \right), \quad \frac{\partial f}{\partial y} = x \left(\frac{x^2 - y^2}{x^2 + y^2} - \frac{4x^2 y^2}{(x^2 + y^2)^2} \right),$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{x^2 - y^2}{x^2 + y^2} \cdot \left(1 + \frac{8x^2 y^2}{(x^2 + y^2)^2} \right),$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{x^2 - y^2}{x^2 + y^2} \cdot \left(1 + \frac{8x^2 y^2}{(x^2 + y^2)^2} \right)$$

bo'ladi.

Aytaylik, $(x, y) = (0, 0)$ bo'lsin. Bu holda funksiyaning hosilalarini ta'rifga ko'ra hisoblaymiz:

$$\frac{\partial f(0,0)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = 0,$$

$$\frac{\partial f(0,0)}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y} = 0,$$

$$\frac{\partial^2 f(0,0)}{\partial y \partial x} = \lim_{\Delta y \rightarrow 0} \frac{\frac{\partial f(0, \Delta y)}{\partial x} - \frac{\partial f(0, 0)}{\partial x}}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{-\Delta y^3}{\Delta y^3} = -1,$$

$$\frac{\partial^2 f(0,0)}{\partial x \partial y} = \lim_{\Delta x \rightarrow 0} \frac{\frac{\partial f(\Delta x, 0)}{\partial y} - \frac{\partial f(0, 0)}{\partial y}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{-\Delta x^3}{\Delta x^3} = 1 \quad \blacktriangleright$$

Yuqorida keltirilgan misollardan ko'rinadiki, $f(x, y)$ funksiyaning

$$\frac{\partial^2 f}{\partial x \partial y}, \quad \frac{\partial^2 f}{\partial y \partial x}$$

aralash hosilalari bir-biriga teng ham bo'lishi mumkin, teng bo'lmasdan qolishi ham mumkin ekan.

2-teorema. Faraz qilaylik, $f(x, y)$ funksiya $(x_0, y_0) \in R^2$ nuqtaning $U_\delta((x_0, y_0))$ atrofida

$$\frac{\partial^2 f(x, y)}{\partial x \partial y}, \quad \frac{\partial^2 f(x, y)}{\partial y \partial x} \quad ((x, y) \in U_\delta((x_0, y_0)))$$

aralash hosilalarga ega bo'lib, bu hosilalar (x_0, y_0) nuqtada uzluksiz bo'lsin. U holda $f(x, y)$ funksiyaning aralash hosilalari (x_0, y_0) nuqtada teng bo'ladi:

$$\frac{\partial^2 f(x_0, y_0)}{\partial y \partial x} = \frac{\partial^2 f(x_0, y_0)}{\partial x \partial y}.$$

◀ Aytaylik, $(x_0 + \Delta x, y_0 + \Delta y), (x_0 + \Delta x, y_0), (x_0, y_0 + \Delta y)$ nuqtalar (x_0, y_0) nuqtaning atrofiga tegishli bo'lsin:

$$(x_0 + \Delta x, y_0 + \Delta y) \in U_\delta((x_0, y_0)), (x_0 + \Delta x, y_0) \in U_\delta((x_0, y_0)), \\ (x_0, y_0 + \Delta y) \in U_\delta((x_0, y_0)).$$

Ushbu

$$\Phi(\Delta x, \Delta y) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) - f(x_0, y_0 + \Delta y) + f(x_0, y_0), \\ \varphi(x) = f(x, y_0 + \Delta y) - f(x, y_0)$$

funksiyalarni qaraylik

Ravshanki,

$$\Phi(\Delta x, \Delta y) = \varphi(x_0 + \Delta x) - \varphi(x_0)$$

bo'ladi. Bu tenglikning o'ng tomoniga Lagranj teoremasini ikki marta qo'llab topamiz:

$$\begin{aligned} \varphi(x_0 + \Delta x) - \varphi(x_0) &= \varphi'(x_0 + \theta_1 \cdot \Delta x) \cdot \Delta x = \\ &= \left[\frac{\partial f(x_0 + \theta_1 \Delta x, y_0 + \Delta y)}{\partial x} - \frac{\partial f(x_0 + \theta_1 \cdot \Delta x, y_0)}{\partial x} \right] \Delta x = \\ &= \frac{\partial^2 f(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y)}{\partial y \partial x} \cdot \Delta x \cdot \Delta y. \quad (0 < \theta_1, \theta_2 < 1) \end{aligned}$$

Shartga ko'ra aralash hosila (x_0, y_0) nuqtada uzluksiz. Demak, $\Delta x \rightarrow 0, \Delta y \rightarrow 0$ da

$$\frac{\partial^2 f(x_0 + \theta_1 \Delta x, y_0 + \theta_2 \Delta y)}{\partial y \partial x} \Delta x \Delta y = \frac{\partial^2 f(x_0, y_0)}{\partial y \partial x} \Delta x \Delta y + o(1)$$

bo'lib,

$$\Phi(\Delta x, \Delta y) = \frac{\partial^2 f(x_0, y_0)}{\partial y \partial x} \Delta x \Delta y + o(1) \quad (7)$$

bo'ladi.

Endi $\Phi(\Delta x, \Delta y)$ funksiya bilan birga quyidagi

$$\psi(y) = f(x_0 + \Delta x, y) - f(x_0, y)$$

funksiyani qaraymiz. Ravshanki,

$$\Phi(\Delta x, \Delta y) = \psi(y_0 + \Delta y) - \psi(y_0)$$

bo'ladi. Yuqoridagidek, bu tenglikning o'ng tomoniga Lagranj teoremasini ikki marta qo'llab, so'ng aralash hosilaning (x_0, y_0) nuqtada uzluksizligidan foydalanib topamiz:

$$\begin{aligned} \psi(y_0 + \Delta y) - \psi(y_0) &= \left[\frac{\partial f(x_0 + \Delta x, y_0 + \theta'_1 \Delta y)}{\partial y} - \frac{\partial f(x_0, y_0 + \theta'_1 \Delta y)}{\partial y} \right] \Delta y = \\ &= \frac{\partial^2 f(x_0 + \theta'_2 \Delta x, y_0 + \theta'_1 \Delta y)}{\partial x \partial y} \Delta x \Delta y = \frac{\partial^2 f(x_0, y_0)}{\partial x \partial y} \Delta x \Delta y + o(1) \end{aligned}$$

$(0 < \theta'_1, \theta'_2 < 1, \Delta x \rightarrow 0, \Delta y \rightarrow 0)$. Demak,

$$\Phi(\Delta x, \Delta y) = \frac{\partial^2 f(x_0, y_0)}{\partial x \partial y} \Delta x \Delta y + o(1) \quad (8)$$

(7) va (8) munosabatlardan

$$\frac{\partial^2 f(x_0, y_0)}{\partial x \partial y} = \frac{\partial^2 f(x_0, y_0)}{\partial y \partial x}$$

bo'lishi kelib chiqadi. ►

Mashqlar

1. Ushbu

$$u = f(x, y), \quad x = t^2 + s^2, \quad y = t \cdot s$$

funksiyaning ikkinchi tartibli differensialini topilsin.

2. Ushbu

$$u = y \cdot \varphi(x^2 - y)$$

funksiya quyidagi

$$\frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} = \frac{u}{y^2}$$

tenglikni qanoatlantirishi isbotlansin.

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